G Vinay Venkata Sai

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Education

Indian Institute of Technology (IIT) Kharagpur 2019 – 2024
Dual Degree (B.Tech + M.Tech)
B.Tech : Electronics & Communication Engineering
M.Tech : Vision and Intelligent Systems (Artificial Intelligence)

Professional Summary

– Markets Data and Regulatory Analytics professional with 2+ years of experience in high-volume, compliance-sensitive financial market environments. Experienced in designing scalable SQL transformations, supporting trade lifecycle analytics, and delivering production-grade reporting solutions for Surveillance, Investigation, and Trade Operations teams. Strong expertise in billion-scale trading datasets, data warehousing principles, reconciliation frameworks, and stakeholder-driven analytics in regulated exchange systems.

Core Competencies

Data & Programming: Advanced SQL (Joins, Window Functions, CTEs, Aggregations, Query Optimization), Hive, Impala, Python (Pandas, NumPy)
Markets & Financial Analytics: Trade Lifecycle Analytics, Order Book Analysis, Impact Cost & Liquidity Metrics, Corporate Actions Adjustments, Equity & Derivatives Markets, Primary & Secondary Market Data Flows, Regulatory Reporting Support
Business Intelligence & Reporting: Tableau, Power BI, Advanced Excel, Automated Dashboard Development
Business & Regulatory Analysis: Requirements Translation (BRD/FRD), Gap Analysis, Root Cause Investigation, UAT Management, Cross-Functional Stakeholder Collaboration

Work Experience

Manager – Data Warehouse & Advanced Analytics | National Stock Exchange (NSE) **June 2024 – Present**

– Collaborated with **Investigation, Surveillance, Trade Operations, and Policy** teams to translate regulatory and trading business requirements into scalable data solutions within a **high-volume exchange environment**.
– Led structured stakeholder workshops to define **KPIs, data definitions, and reporting standards**, reducing requirement clarification cycles by 35% and improving delivery efficiency.
– Performed cross-platform data analysis across **Equity, Derivatives, RFQ, OFS, IPO, and secondary market systems**, identifying data quality gaps and strengthening compliance-sensitive reporting accuracy
– Designed and optimized complex **SQL- and Python-based analytics frameworks** supporting 10+ high-priority regulatory and operational reporting use cases monthly on billion-scale trading datasets
– Conducted production-grade **root cause analysis** and managed **UAT validation cycles** to ensure **data integrity, audit readiness, and post-deployment stability** across high-volume trading data platforms.

Key Analytics Projects

Multi-Broker Portfolio Consolidation & Business Intelligence **NSE Application**

– Designed consolidated portfolio analytics across multiple broker feeds, defining **corporate action adjustment logic** (splits, dividends, bonuses), cost-basis treatment, and 1-day, realized, and unrealized **P&L computation** for equity instruments.
– Translated complex financial computation frameworks into technical specifications, ensuring accurate trade aggregation, portfolio valuation, and client-level reporting consistency.

- Built automated **reconciliation and validation controls** resolving 40+ structural data discrepancies, achieving near-100% portfolio data accuracy.
- Delivered a **production-grade portfolio module** within the NSE trading ecosystem, enabling **pan-India investors** to monitor consolidated equity holdings across brokers in a unified interface.

Impact Cost Analysis & Performance Optimization **Trading Analytics**

- Computed **impact cost and liquidity metrics** across **1B+ daily equity order records** during live trading hours, supporting exchange-level liquidity monitoring and regulatory analytics.
- Re-architected legacy file-based batch processing into a direct **database-connected analytics pipeline**, eliminating file I/O dependencies and improving system reliability.
- Implemented **multiprocessing-based Python computation** and optimized SQL transformations, reducing execution timelines by 80% and enabling intraday operational reporting.
- Built structured database output layers and **Tableau dashboards** to support interval-based trading performance analysis and compliance-grade reporting.

Client Onboarding Analytics & Business Process Improvement **Customer Analytics**

- Performed requirements gathering sessions with business stakeholders to understand client onboarding challenges and defined business rules for client data reconciliation and validation.
- Designed SQL-based data quality checks and reconciliation logic for **500K+ client records**, improving data reliability and regulatory readiness by 30%.
- Implemented predictive analytics model to identify first-time clients with **98% accuracy**, enabling targeted marketing campaigns that improved outreach effectiveness by **15–20%**.
- Documented business processes and created process flow diagrams to support knowledge transfer and continuous improvement initiatives.

Business Process Automation & Operational Analytics **Process Improvement**

- Conducted gap analysis and stakeholder interviews to identify manual reporting pain points and defined requirements for automated operational tracking solutions.
- Developed automated reporting dashboards using Python and Tableau, replacing manual Excel-based processes and improving data accuracy and timeliness.
- Reduced manual reporting effort by **30%** and issue resolution turnaround time by **25–30%**, enabling faster business decision-making and operational efficiency gains.
- Created comprehensive documentation including business requirements documents (BRD) and functional specifications for deployed automation solutions.